Square or Cubic? Finding the Unit Mistake

Answer Key

Grade 4–5

Quick Answers

1. 40	3. 216	5. 100	7. 60
2. 120	4. 96	6. 1000	8. —

Curriculum

Level: Grade 4–5
4.MD / 5.MD – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

2. *If they got 40: multiplied only the two base edges, so the answer counts flat squares instead of cubes*
3. *If they got 36: squared the edge instead of cubing it, so the number counts squares while the label says cubes*
4. *If they got 64: cubed the edge, computing the volume and labelling it as surface area*
5. *If they got 10: used the one-dimensional factor once, as if a square unit converted like a length*
6. *If they got 10: used the one-dimensional factor once, as if a cubic unit converted like a length*
6. *If they got 100: squared the factor, treating a cubic unit as if it were a square unit*
7. *If they got 120: multiplied in the depth as well, computing a volume and labelling the result in square meters*

The one question behind this whole sheet: *how many lengths were multiplied?* One length is a plain unit, two lengths make square units, three lengths make cubic units. Every mistake on this page is a number built from the wrong count of lengths, or a right number wearing the wrong label.

1. Area of a sticker 8 cm by 5 cm.

Area covers a flat surface, so lay unit squares over it: 5 rows of 8 tiles. That is two lengths multiplied, and two lengths always give *square* units.
$8 \times 5 = 40$, so the area is $\boxed{40 \text{ cm}^2}$.

2. Volume of a box 8 cm by 5 cm by 3 cm.

Volume fills a space, so pack unit cubes inside. One layer on the floor of the box holds $8 \times 5 = 40$ cubes. The box is 3 centimeters tall, so it holds 3 such layers.
$40 \times 3 = 120$ — three lengths multiplied, so the unit is *cubic*: $\boxed{120 \text{ cm}^3}$.
Watch for: stopping at 40. That is one flat layer, an area, not the volume.

3. Find the mistake: Dev wrote $V = 6 \times 6 = 36$ cubic centimeters.
(a) Dev multiplied only *two* edges. Two lengths measure a flat face, so 36 is the area of one face of the cube, in square centimeters — he squared the edge and then wrote a cubic label on it. A cube has three edges meeting at a corner, and all three must be multiplied.
(b) Correct work: one layer of the cube holds $6 \times 6 = 36$ unit cubes, and the cube is 6 centimeters tall, so it holds 6 layers.
$36 \times 6 = 216$, so the volume is $\boxed{216\ \text{cm}^3}$.

4. Surface area of a cube with 4-centimeter edges.
Surface area is the area of the outside, so it is built from flat faces — square units, not cubic ones. One face is a square with 4-centimeter sides: $4 \times 4 = 16$ square centimeters. A cube has six identical faces.
$6 \times 16 = 96$, so the surface area is $\boxed{96\ \text{cm}^2}$.
Watch for: $4 \times 4 \times 4 = 64$. That is the volume; it answers a different question and carries cubic units.

5. How many square millimeters cover one square centimeter?
A square centimeter is a square 1 centimeter on each side. Each side is 10 millimeters long, so the square holds 10 rows of 10 small squares.
$10 \times 10 = 100$, so one square centimeter is $\boxed{100\ \text{mm}^2}$.
The rule to say out loud: for *square* units the length factor gets used twice — squared.

6. Find the mistake: Ana said 10, her partner said 100.
(a) Ana used the length factor *once*: 10 millimeters per centimeter is a rule about *lengths*, and she applied it to a solid. Her partner used the factor *twice*, which is the rule for flat square units — correct for square millimeters, but this is a cubic unit. A cubic centimeter is a cube 1 centimeter on every edge, so all three edges must be converted.
(b) Each edge is 10 millimeters, so the cube holds 10 layers of 10 rows of 10 small cubes: $10 \times 10 \times 10$.
That gives $\boxed{1000\ \text{mm}^3}$ in one cubic centimeter.

7. Find the mistake: Kai wrote $12 \times 5 \times 2 = 120$ square meters.
(a) Two things are wrong, and they are linked. First, Kai multiplied *three* lengths — length, width and depth — which measures the water the pool holds, not the floor. Second, he labelled that three-length answer in square meters; three lengths always give cubic units, so the number and the label disagree even inside his own work. His 120 is really 120 cubic meters of water.
(b) The floor is a flat rectangle, so use only the two lengths that lie in the floor: 12 meters by 5 meters.
$12 \times 5 = 60$, so the cover must be $\boxed{60\ \text{m}^2}$.
Watch for: an answer of 120 with any label. The depth of 2 meters is a distractor for this question.

8. "Area and volume are the same thing." (open response — read the drawing and the reasoning)

The classmate is wrong, and the drawing is the proof. Covering a flat rectangle uses tiles, and each tile is one square unit; the number of tiles is (rows)×(tiles per row), which is two lengths multiplied. Filling a box uses blocks, and each block is one cubic unit; the number of blocks is (layers)×(rows per layer)×(blocks per row), which is three lengths multiplied. The unit records how many directions were measured, so it cannot be chosen freely at the end.

$\boxed{\text{Area uses two lengths; volume uses three}}$

Full credit shows tiles filling a flat shape *and* cubes filling a solid, and says in words that two lengths give square units while three give cubic units. *Not full credit:* "area is flat and volume is 3D" with no link to the multiplying, or a drawing with no sentence.